

Open Architecture and Modular Paradigm of Cyclus

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Outline

1 Top Level Simulators

Introduction

Lessons Learned

2 Cyclus Modular Architecture

Model Hierarchy

Modularity

Example : Market-Facility Interface

Extensibility

3 Cyclus Development Paradigm

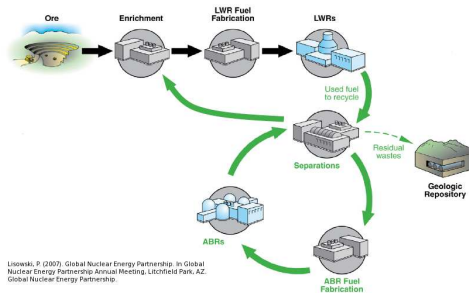
' Modified Open' Source

Quality Control

Testing Framework



Top Level Simulators



Lisowski, P. (2007). Global Nuclear Energy Partnership. In Global Nuclear Energy Partnership Annual Meeting, Litchfield Park, AZ. Global Nuclear Energy Partnership.

Figure: Top level simulators are intended to model the collective behavior of various fuel cycle decisions and strategies.[3]



Lessons Learned

Areas of unmet need in previous simulators include:

- **Usability** directed toward a wide range of user sophistication.
- **Performance** solving simple problems in interactive time scales
- **Fidelity** accommodating a range of levels of detail commensurate with user sophistication

Valuable software development practices have been identified [2][1]:

- Modularity allows the core infrastructure to be independent of proprietary and/or sensitive data and/or models.
- Extensibility with a focus on both robustness and flexibility allows for myriad potential developer extensions.
- Openness supports collaboration, verification, validation, and code sustainability.



Cyclus Modular Architecture and Open Development

The combination of modular encapsulation within the software architecture and an open development paradigm allows for collaboration at multiple levels of simulation detail and data security.



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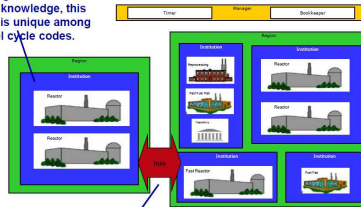
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Region Institution Facility Hierarchy

Institutions represent companies and government entities. To our knowledge, this modeling layer is unique among comparable fuel cycle codes.



User-specified rules might state whether one fuel cycle state can send materials to another, or define some special fuel-trade contract.

3

Figure: The Region, Institution, Facility Hierarchy of the Cyclus paradigm was a success of the GENIUSv2 code.[4]



Encapsulation

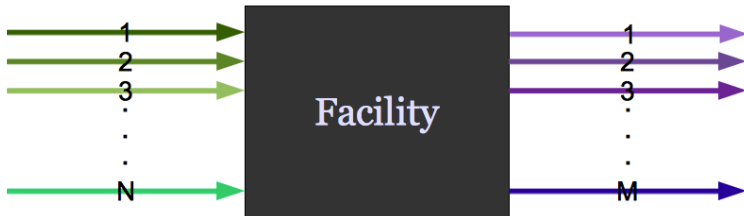


Figure: Regions, Institutions, Facilities, and Markets are all black boxes.

Module Interfaces

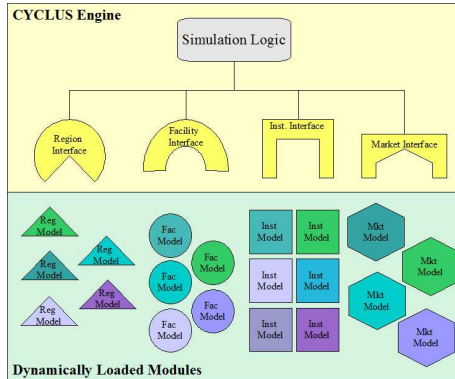


Figure: Well defined model interfaces facilitate model interchange. The user may choose the model at their desired level of detail.



Facilities Are Black Boxes

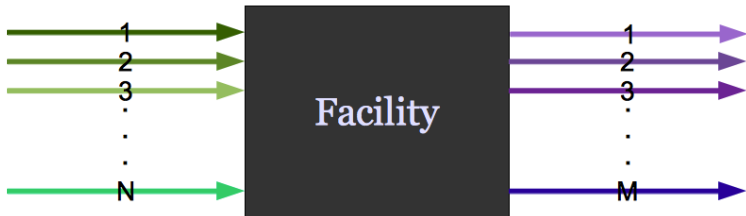


Figure: Each facility in the simulation makes requests and offers to fill its stocks and empty its inventory respectively.



Facilities Are Black Boxes



Figure: A facility might only make offers.



Facilities Are Black Boxes

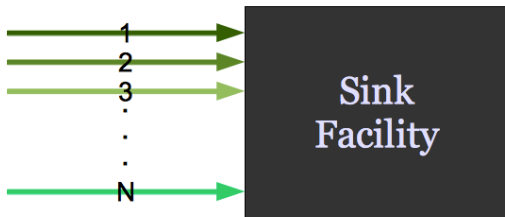


Figure: A facility might only make requests.



Each Commodity is Associated with a Market



Figure: A market receives offers and requests concerning its commodity.



The Market Solves the Matching Problem

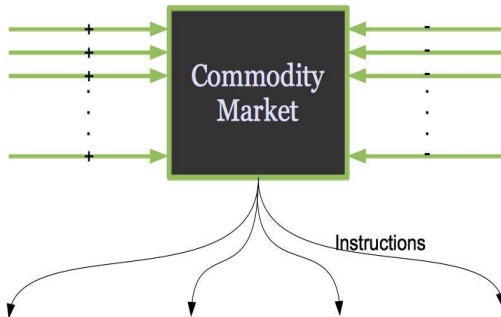


Figure: When the Market's arbitrary algorithm solves the matching problem, the Market sends instructions to the offering facilities.



A Simple Example

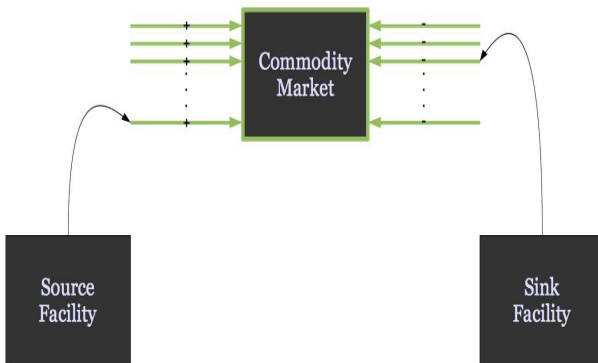


Figure: The source sends an offer and the sink sends a request.



A Simple Example

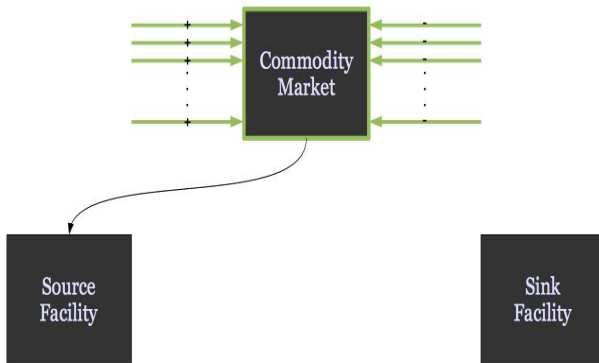


Figure: The Market solves the problem and instructs the source facility to send a certain amount to the sink facility.



A Simple Example

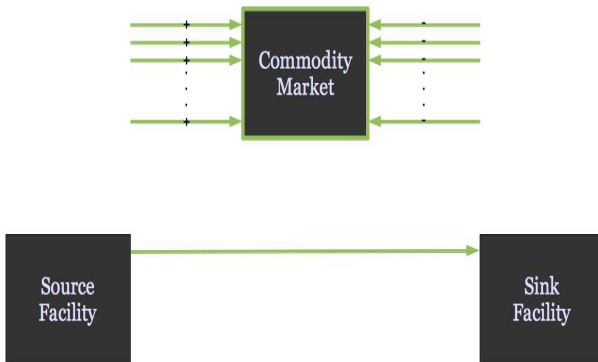


Figure: The source facility sends the material directly to the sink facility.



This Market Model Scales for Complex Systems

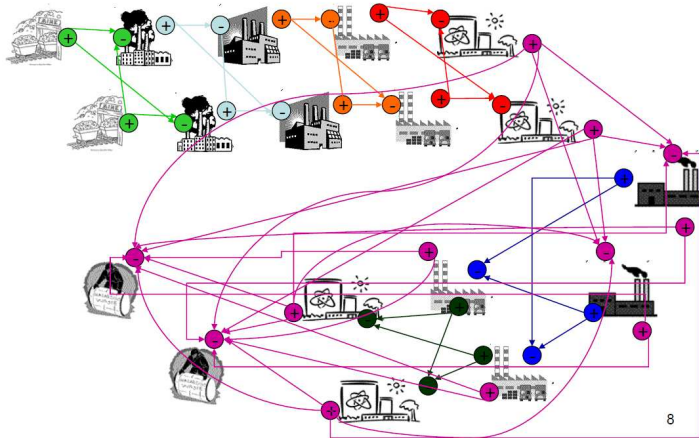


Figure: Well designed interfaces and strict encapsulation support scalability of the Market-based simulation paradigm [4]



Dynamic Module Loading : User

With a dynamic, plug-in implementation, the simulation logic is independent of the available models and models are loaded as shared libraries at runtime.

	input.xml
	<pre><simulation> <startYear>1962</startYear> <duration>1200</duration> <region> <name>UChicago</name> <DeployRegionModel> <deployment> <facility> <name>ChiPile1</name> <model>RecipeReactor</model> </facility> <year>1942</year> </deployment> </DeployRegionModel> . . . </region> . . . </simulation></pre>

Figure: XML input parsing and a relaxNG schema provide a simplified XML interface



Dynamic Module Loading : Developer

With a dynamic, plug-in implementation, the simulation logic is independent of the available models and models are loaded as shared libraries at runtime.

	Model.cpp
	<pre> #include Model.hpp #include dlfcn.h . . . mdl_ctr* loadModel(name){ void* model = dlopen(name.c_str(), RTD_LAZY) mdl_ctr* new_model = (mdl_ctr*) dlsym(model,"construct"); . . return new_model; } </pre>
	RecipeReactor.cpp
	<pre> #include RecipeReactor.hpp . . . extern "C" Model* construct() { return new RecipeReactor(); } . . . </pre>

Figure: Dynamic c library loading separates simulation logic from knowledge of



Cross Platform, Multi-language

CMake (Crossplatform Make) build system is cross platform. Thus, users may build and run Cyclus on Windows, Mac, or Linux operating systems. CMake is also multilingual, so modules may be written in any language or tool that can be made into a shared object library or wrapped with C++.

These include (but are not limited to):

- Fortran
- Python
- R
- Java
- Lisp
- IDL
- Matlab
- Perl
- etc. . .



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Open Source Repository

This open source repository provides a centralized location for documentation, developer history, and unhindered developer access.

code.google.com/p/cyclus/source/browse/#svn%2Ftrunk%2Fsrc

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Source path: svn/ r339 r340

Directories	Filename	Size	Rev	Date	Author
*svn	App.cpp	2.0 KB	r314	May 19, 2011	kathyhuff
branches	CMakeLists.txt	3.5 KB	r289	Apr 30, 2011	kathyhuff
doc	Commodity.cpp	903 bytes	r111	Jul 24, 2010	kathyhuff
trunk	Commodity.h	2.3 KB	r111	Jul 24, 2010	kathyhuff
input	Communicator.cpp	1.8 KB	r117	Jul 29, 2010	kathyhuff
src	Communicator.h	1.6 KB	r117	Jul 29, 2010	kathyhuff
CMake	InputXML.cpp	7.0 KB	r301	May 5, 2011	kathyhuff
Models	InputXML.h	9.2 KB	r115	Jul 28, 2010	kathyhuff
Testing	Logician.cpp	8.4 KB	r240	Feb 23, 2011	Matthew.Gidden
Utility	Logician.h	7.2 KB	r166	Oct 19, 2010	kathyhuff
doc	Material.cpp	22.8 KB	r334	Jun 4, 2011	Matthew.Gidden
wiki	Material.h	14.5 KB	r334	Jun 4, 2011	Matthew.Gidden

Your project is using approximately 7.0 MB out of 4096 MB total quota.
You can [reset this repository](#) so that svnSync can be used to upload existing code history.



'Modified Open' Source

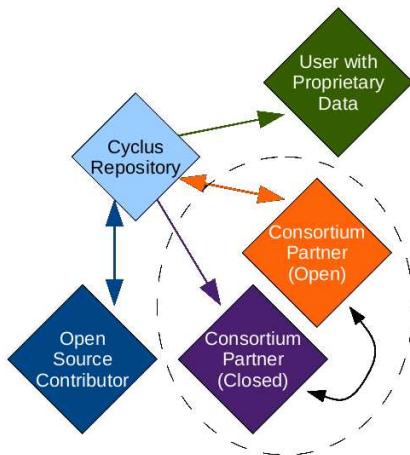


Figure: License, architecture, and development paradigm allow varying levels of code sharing and data security.



Version Control

This open source repository employs a version control system for provenance, developer acces, and reproducibility of results.

code.google.com/p/cyclus/source/browse/#svn%2Ftrunk%2Fsrc

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> trunk	Commodity.h	2.3 KB	r111	Jul 24, 2010	kathyhuff
> input	Communicator.cpp	1.8 KB	r117	Jul 29, 2010	kathyhuff
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Issue Tracking

A built-in issue tracking system allows ticketing of development milestones and public bug reporting.

code.google.com/p/cyclus/issues/list

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katyhuff@gmail.com

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New issue Search Open issues for Search Advanced search Search tips

Tip: Type ? for issue tracker keyboard shortcut help.

Select: All None Actions...

ID	Type	Status	Priority	Milestone	Owner	Summary + Labels
☐ 42	Enhancement	Assigned	High	NullSimulation	Matthew.Gidden	Possible Issue Passing Material During Market Matches
☐ 21	Enhancement	Accepted	High	INPRO_01	Matthew.Gidden	Growth region - on the fly
☐ 22	Enhancement	Accepted	High	INPRO_01	Matthew.Gidden	Flesh out INPRO required pre-processing
☐ 23	Enhancement	Accepted	High	INPRO_01	Matthew.Gidden	Flesh out INPRO required post-processing
☐ 25	Enhancement	Accepted	High	INPRO_01	Matthew.Gidden	Determine the remaining capabilities for INPRO after deployment, pre-, and post-processing
☐ 28	Enhancement	Accepted	High	INPRO_01	katyhuff	Output Database - Target Date 4/4/2011
☐ 9	Design	Started	High	INPRO_01	katyhuff	Define Fundamental Data Unit for Output
☐ 38	----	Accepted	Low	INPRO_01	Matthew.Gidden	Add XML input for BuildRegion Vector Capability
☐ 41	----	Accepted	Critical	NWTRB_01	Matthew.Gidden	Document and complete NWTRB simple test case
☐ 26	Enhancement	New	High	NWTRB_01	Royal.Elmore	Create NWTRB Facility Catalog - Target Date 4/11/2011
☐ 27	Enhancement	New	High	NWTRB_01	Royal.Elmore	NWTRB Recipe Book - Target Date 4/11/2011



Collaborative Verification

The **transparency** inherent in this type of open source development path also **facilitates code review** by exposing available content to verification and validation **by collaborators** with diverse areas of specialization and levels of expertise.

code.google.com/p/cyclus/

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nuclear, energy, simulation, university, python, c, fuel.cycle

Members
katyhuff, uwgonuke, code.scrivener, 4 committers

Your role
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Links

External links
cnerg

Groups
Cyclus Developers List

As a successor to **GENIUS**, the next-generation fuel cycle systems analysis simulation tool, **Cyclus** will preserve many a software architecture that provides a great deal of flexibility, both in terms of modifying the underlying modeling algorithm different levels of complexity to different users.

The Cyclus modeling paradigm will let users reconfigure the basic building blocks of a simulation without changing the a of a simulation will be a commodity market that collects offers and requests and matches them according to some algo able to select which type of algorithm is used for each market by selecting a MarketModel and configuring it with a parti defined by that MarketModel. Changing the parameters of a market will change its performance and selecting a different completely change its behavior.

- [Cyclus Introduction](#)
- [Getting and Building Cyclus](#)
- [User Documentation](#)
- [Developer Documentation](#)

Internal code development

Most internal code development at this time will be led by [Katy Huff](#), [Paul Wilson](#), [Matthew Gidden](#), and [Royal Elmore](#). group. Near term goals include :

- replicating GENIUSv2 functionality
- developing a unit test suite
- developing a library of basic data and a catalog of market and facility configurations.

External code development

Interested developers are welcome and encouraged to contribute, but will experience significant code instability in the a



Testing Framework

A testing framework built on a cross-platform, multi-language build system (CMake) allows developers to incorporate unit and integration tests into their code before it is committed.



References I

- [1] Kathryn Huff and Brent Dixon.
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Technical Report fcrd-sysa-2010-000110, July 2010.
- [2] Kathryn Huff, Paul P. H. Wilson, and Matthew Gidden.
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In *Global Nuclear Energy Partnership Annual Meeting*, 2007.
- [4] K. M Oliver.
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